[3]:
```
import pandas as pd
import pandas as pd
import matplotlib.pyplot as plt
```

[4]:
```
import numpy as np

x = np.array([50,55,60,70,75,80,90,100])
y = np.array([75,50,90,55,100,60,80,70])
```

[6]:
```
model = np.polyfit(x,y,1)
model
```

[6] : array([0.0952381, 65.5952381])

[7] :
```
predict = np.poly1d(model)
predict(59)
```

[7]: 71.21428571428571

[8] :
```
y_pred = predict(x)
y_pred
```

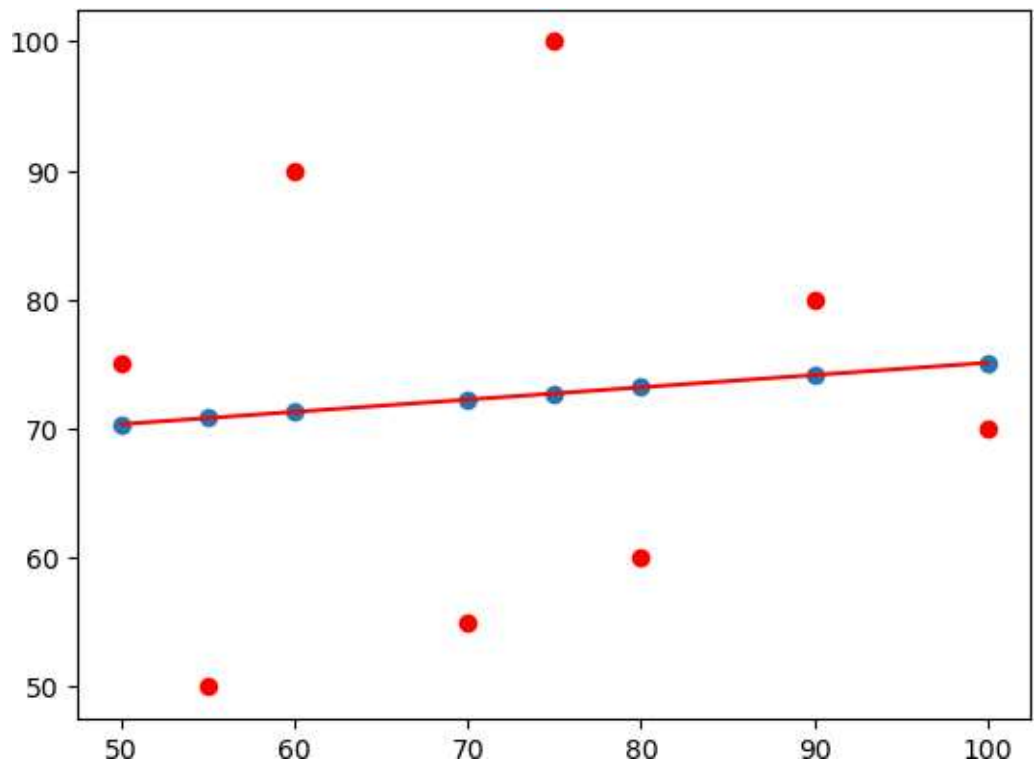
[8] : array([70.35714286, 70.83333333, 71.30952381, 72.26190476, 72.73809524,
73.21428571, 74.16666667, 75.11904762])

[9] :
```
from sklearn.metrics import r2_score
r2_score(y, y_pred)
```

[9]: 0.009070294784580657

[10] :
```
y_line = model[1] + model[0] * x
plt.plot(x,y_line, c = 'r')
plt.scatter(x,y_pred)
plt.scatter(x,y, c = 'r')
```

[10] : <matplotlib.collections.PathCollection at 0x1b8a06956d0>



[11] :
```
from sklearn.datasets import fetch_california_housing
import pandas as pd
import numpy as np
```

[12] :
```
california = fetch_california_housing()
```

[13] :
```
df = pd.DataFrame(california.data, columns=california.feature_names)
```

[14] :
```
df['Target'] = california.target
```

[15] :
```
print(df.head())
```

```
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude  Target
0    -122.23   4.526
1    -122.22   3.585
2    -122.24   3.521
```

```
3     -122.25    3.413
4     -122.25    3.422
```

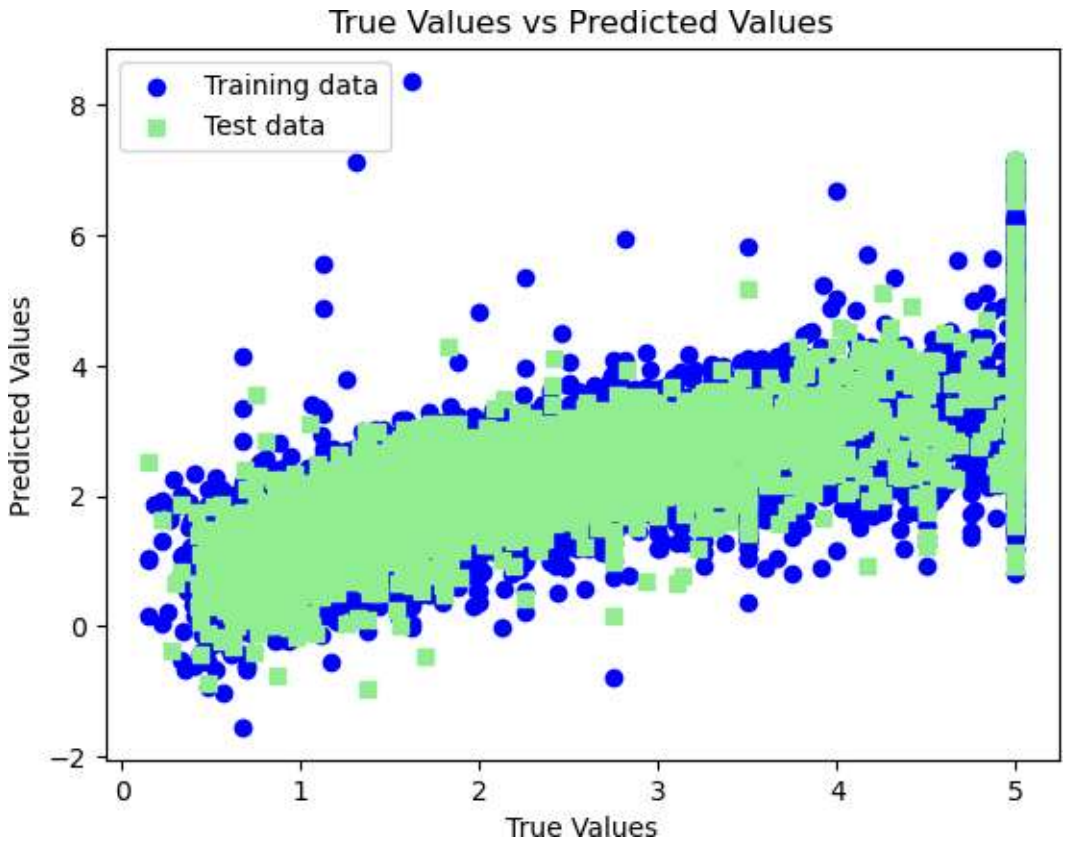
[16] :
```python
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   MedInc      20640 non-null  float64
 1   HouseAge    20640 non-null  float64
 2   AveRooms    20640 non-null  float64
 3   AveBedrms   20640 non-null  float64
 4   Population  20640 non-null  float64
 5   AveOccup    20640 non-null  float64
 6   Latitude    20640 non-null  float64
 7   Longitude   20640 non-null  float64
 8   Target      20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
None
```

[17] :
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt


from sklearn.datasets import fetch_california_housing
california = fetch_california_housing()


data = pd.DataFrame(california.data)


data.columns = california.feature_names
data.head()


data['PRICE'] = california.target


print(data.isnull().sum())

 x = data.drop(['PRICE'], axis=1)
y = data['PRICE']

model_selection import train_test_split
```

```
xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size=0.2,␣
↪random_state=0)


lm = LinearRegression()
model = lm.fit(xtrain, ytrain)


ytrain_pred = lm.predict(xtrain)
ytest_pred = lm.predict(xtest)


df_train = pd.DataFrame({'True Values': ytrain, 'Predicted Values': ytrain_pred})
df_test = pd.DataFrame({'True Values': ytest, 'Predicted Values': ytest_pred})


from sklearn.metrics import mean_squared_error
mse_train = mean_squared_error(ytrain, ytrain_pred)
mse_test = mean_squared_error(ytest, ytest_pred)


print(f'Mean Squared Error (Training data): {mse_train}')
print(f'Mean Squared Error (Testing data): {mse_test}')


plt.scatter(ytrain, ytrain_pred, c='blue', marker='o', label='Training data')
plt.scatter(ytest, ytest_pred, c='lightgreen', marker='s', label='Test data')
plt.xlabel('True Values')
plt.ylabel('Predicted Values')
plt.title("True Values vs Predicted Values")
plt.legend(loc='upper left')
plt.plot()
plt.show()
```

```
MedInc        0
HouseAge      0
AveRooms      0
AveBedrms     0
Population    0
AveOccup      0
Latitude      0
Longitude     0
PRICE         0
dtype: int64
Mean Squared Error (Training data): 0.5234413607125448
Mean Squared Error (Testing data): 0.5289841670367192
```


True Values vs Predicted Values
Training data
Test data
Predicted Values
True Values
8
6
4
2
0
−2
0
1
2
3
4
5

[ ]: